

Neurological birth defects

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Session Objectives

1. Review the key embryological events of the nervous system and apply them to congenital neurological defects
2. Describe and discuss the most common **Neural tube defects** (e.g. spina bifida)
3. Describe and discuss the most **common forebrain (e.g.; holoprosencephaly, lissencephaly) and posterior fossa (e.g.; Chiari and Dandy Walker) anomalies**
4. Describe and discuss **syringomyelia and hydrocephalus**
5. Describe **pre and perinatal brain injuries**

Topics today

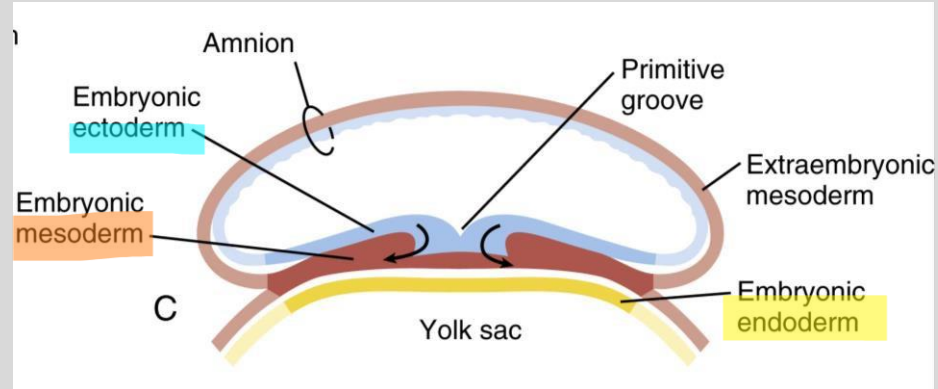
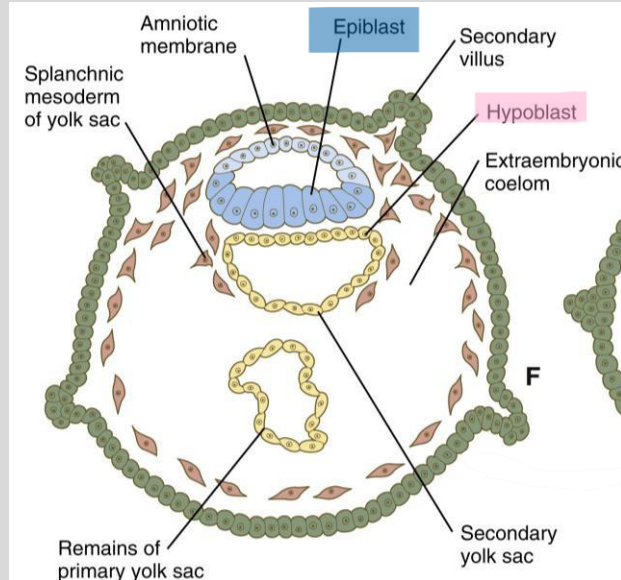
- **Neuroembryology review**
 - Neural tube defects**
 - Spina bifida
 - Myelomenigocele
 - Meningocele
 - Encephalocele
 - Anencephaly
 - Disorders of neuronal migration**
 - Lissencephaly
 - Schizencephaly
 - Neuronal heterotopias
 - Polymicrogyrias
- **Agenesis/dysgenesis disorders**
 - Corpus callosum agenesis
 - Holoprosencephaly
 - Microcephaly
 - Megalencephaly
- **Posterior fossa anomalies**
 - Chiari malformations
 - Dandy-Walker
- **Involving CSF**
 - Syringomyelia
 - Hydrocephalus
- **Cerebral palsy**

Neuroembryology review

- Gastrulation (day 17 of gestation) Epiblast cells from bilaminar disc (epiblast and hypoblast) migrate through primitive *groove*, streak & node, developing the trilaminar disc (ectoderm, endoderm, mesoderm)

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- Trilaminar disc

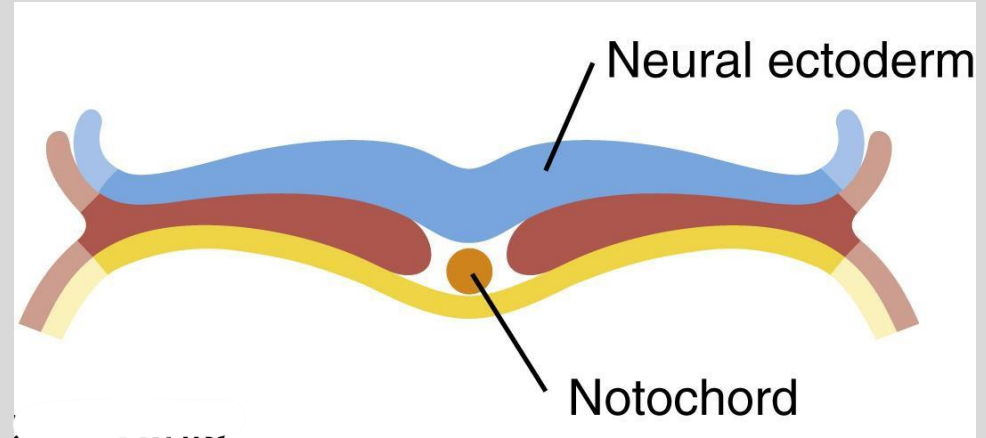
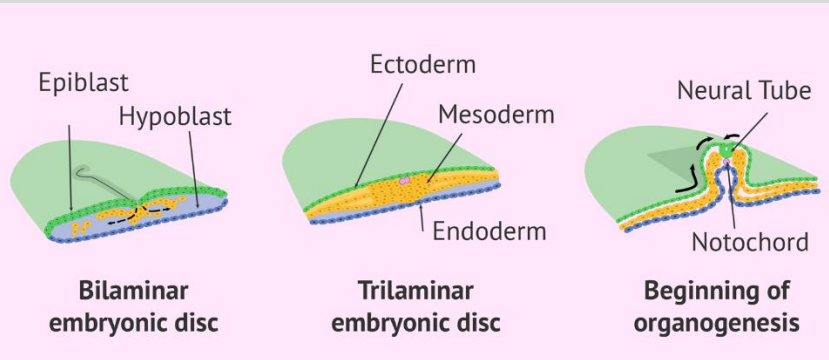
- Bilaminar disc

Neuroembryology

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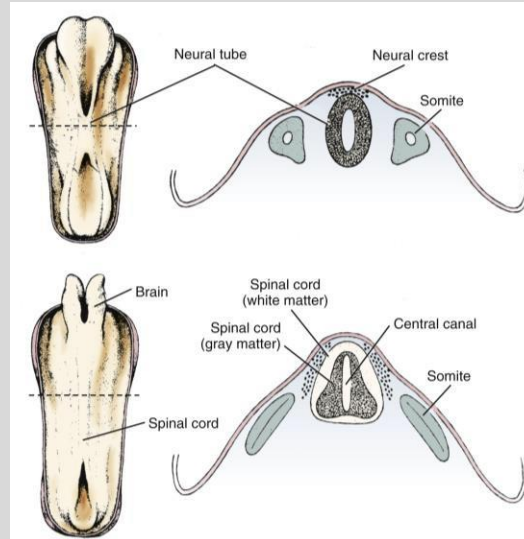
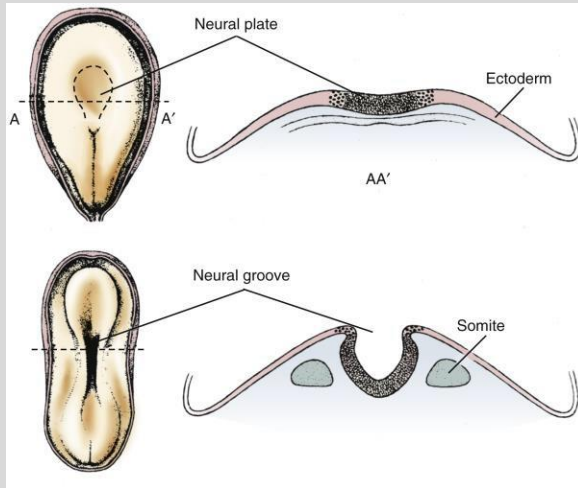
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- Notogenesis (week 3-4 of gestation)
 - Notochord forms between ectoderm and endoderm from epiblast cells, essential for development of nervous system
 - Notochord degenerates after embryogenesis, remnants become the nucleus pulposus of intervertebral discs



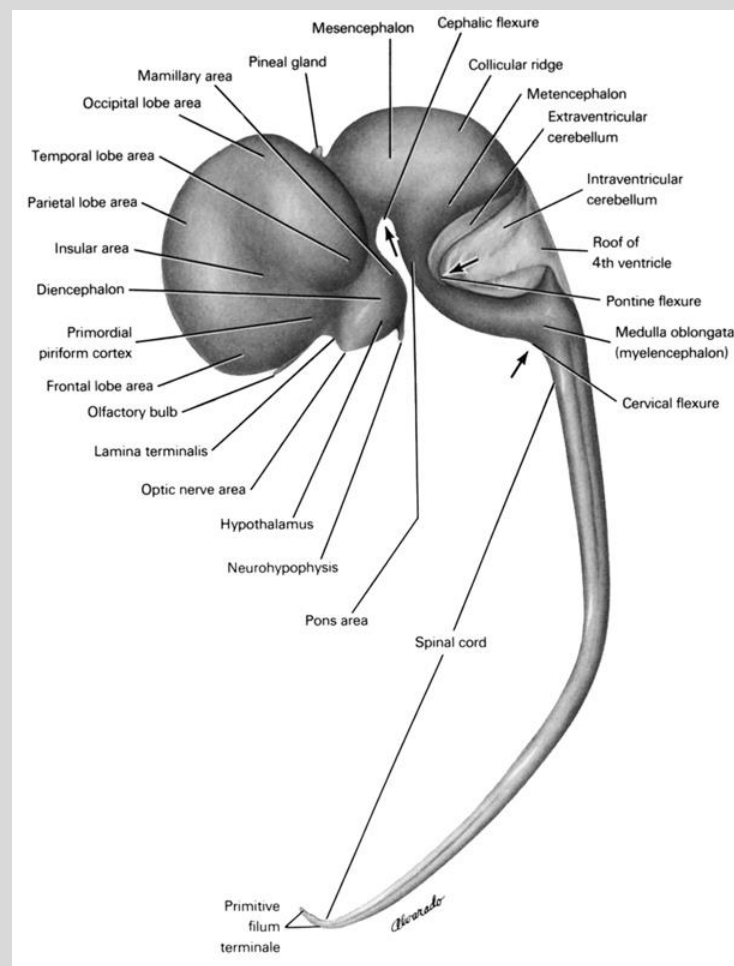
Neural induction and neurulation

- Notochord induces (secretes signals to) the ectoderm & ectoderm thickens - becomes neural plate (neurulation) narrows and elongates; and cranial/caudal neuropores fold into neural tube - precursor to CNS
- Neural plate's edges elevate to form neural folds, creating central depression called neural groove
- Neural folds move towards each other and fuse midline forming neural tube
- Somites, blocks of mesoderm, form alongside developing neural tube (give rise to vertebrae, ribs & muscles) – 3rd week



Segmentation

- Along the neural tube different segments differentiate into primitive structures
- Different areas become different parts of the nervous system



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- https://www.ehd.org/virtual-human-embryo/aohe_figure.php?chapter=8&figure=1b
- ~ 9-10 weeks of gestation

Further growth and maturation...

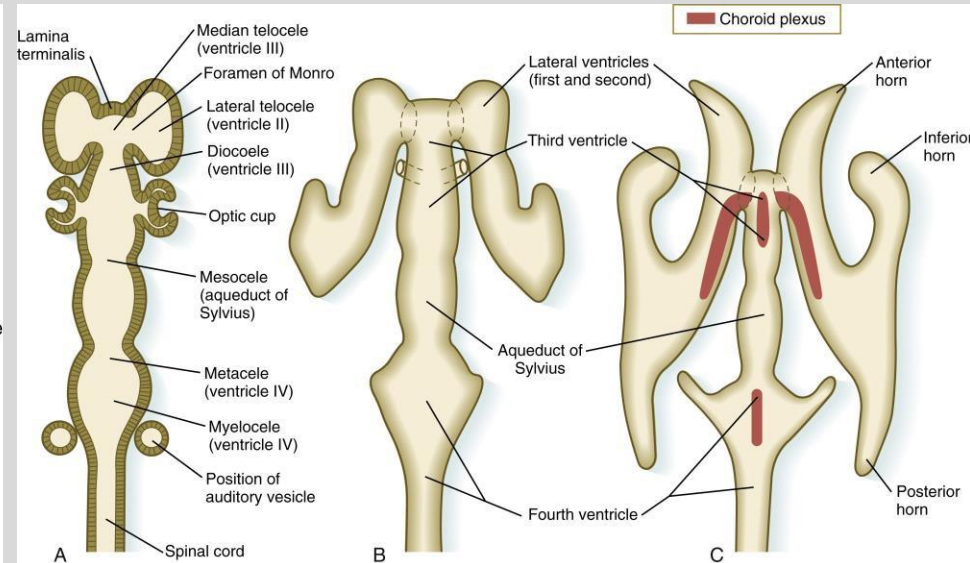
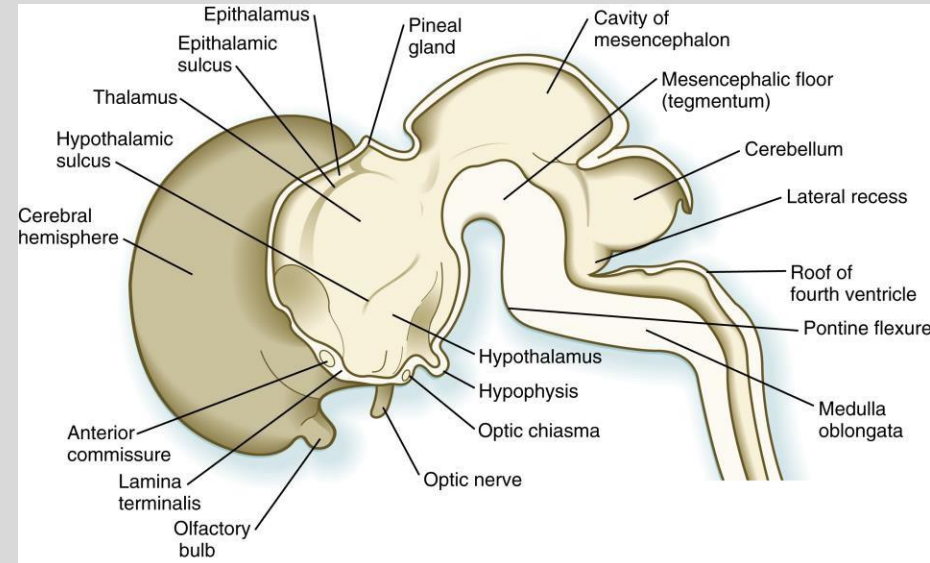
- ~9 wks.
- Notice how structures grow *around* the neural tube (nuclei, cortex, nerves)
- Eventually tube *itself* becomes the inside of the *ventricular system* and spinal-ependymal canal

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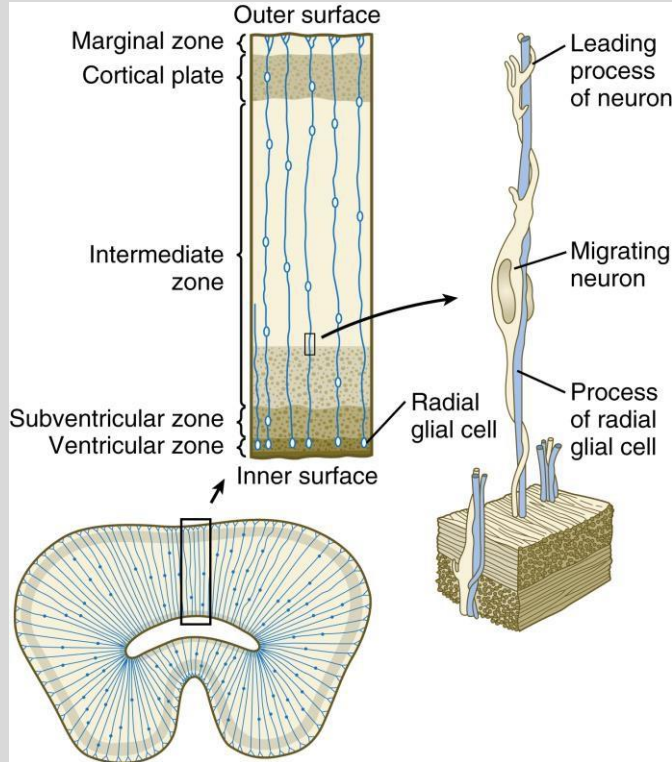
Neural tube &
brain vesicles
~4-6 wks.

Ventricular system, developing left
& postnatal right



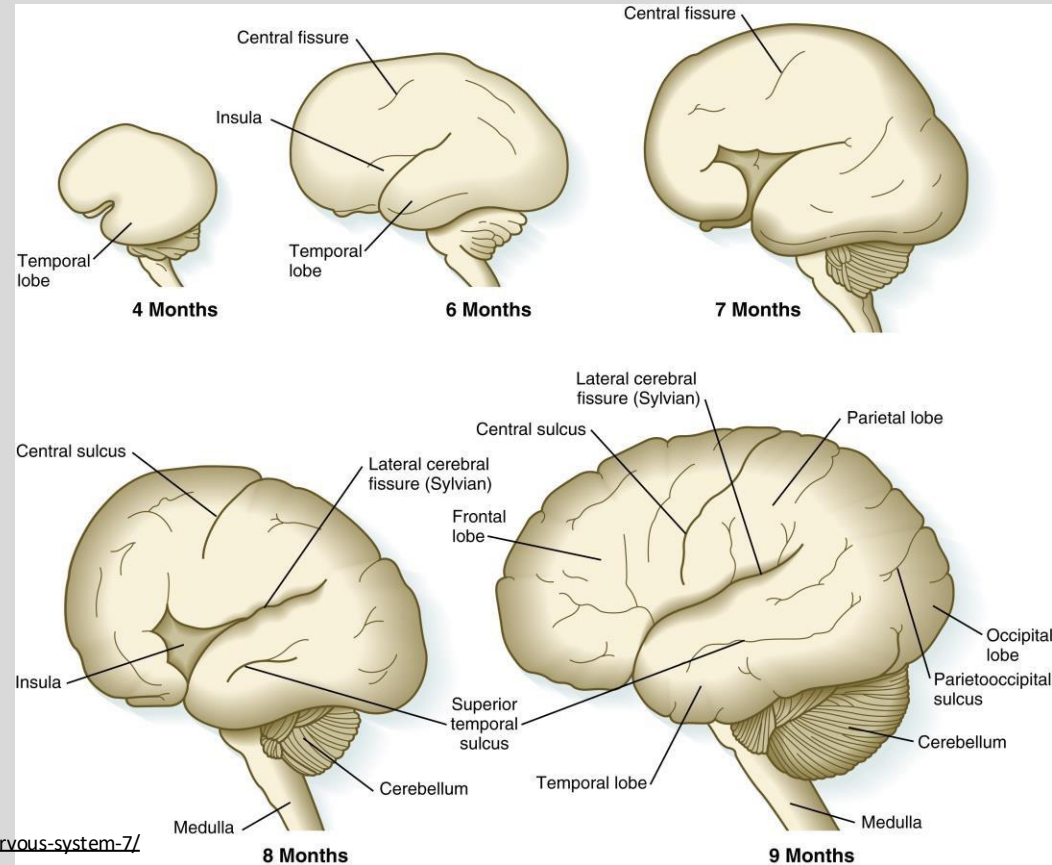
Neuronal migration

- Nervous system & cortex continue to mature via neuronal migration (6-layer cerebral cortex, cerebellum has 3), and brain takes shape



<https://clinicalgate.com/nervous-system-7/>

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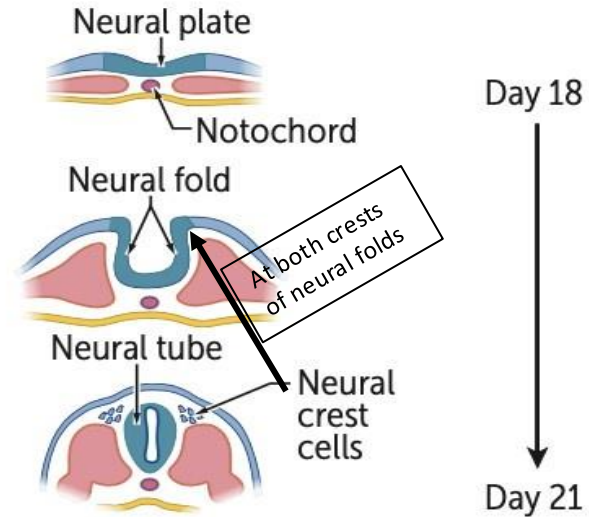
Neuroembryology from First Aid

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- Notochord (nucleus pulposus precursor) induces ectoderm to make neuroectoderm
- Neural plate produces neural tube & crest cells which migrate to form the PNS
neurons & glial cells - neuroepithelia in neural tube form CNS neurons & glial cells

Neural development

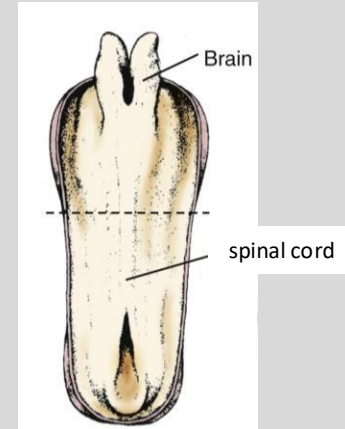


Neural tube defects

- Failure of neural tube/ neuropores to close around **week 3-4** → persistent connection between amniotic cavity and spinal canal
- Maternal serum/amniotic **alpha-fetoprotein** ↑ (except **spina bifida occulta** → **normal** AFP), and **amniotic acetylcholinesterase** ↑ (week 16-18 sampling)
 - Many risk factors: Drugs (**valproic acid, carbamazepine, TMP-SMX, phenytoin, phenobarbital, primidone**), **maternal malnutrition, alcoholism, low folate levels, maternal obesity/diabetes**, and **genetic** determinants (ex. mutations in folate-responsive or folate- dependent enzyme pathways)
 - **Folate** insufficiency is associated with **NTDs**: Current recommendations, potentially become pregnant, consume at least 0.4 mg/day of folic acid
 - **Maternal diabetes** significantly increases risk of NTDs
- Comprise most CNS congenital anomalies and include (covered here) **spina bifida, meningocele, myelomeningocele, encephalocele, and anencephaly**

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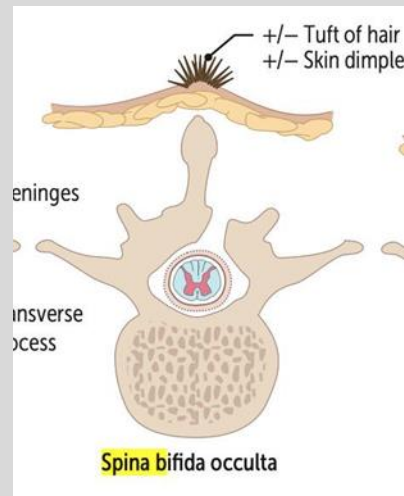


Youmans and Winn
Neurological Surgery 8th
Ed.



Spina bifida occulta

- **Closed** neural tube defect - midline fusion defect of vertebral bodies **without protrusion** of meninges or spinal cord
- Mostly neurologically **asymptomatic**
- **Cutaneous manifestations** are common on the **lower back**
 - A, Midline sacral **hemangioma**
 - B, **Capillary malformation** with **hypertrichosis** (excessive hair)
 - C, Human **tail**
 - D, Midline area of hypertrichosis (**faun tail**)
 - E, Lumbosacral **lipoma**



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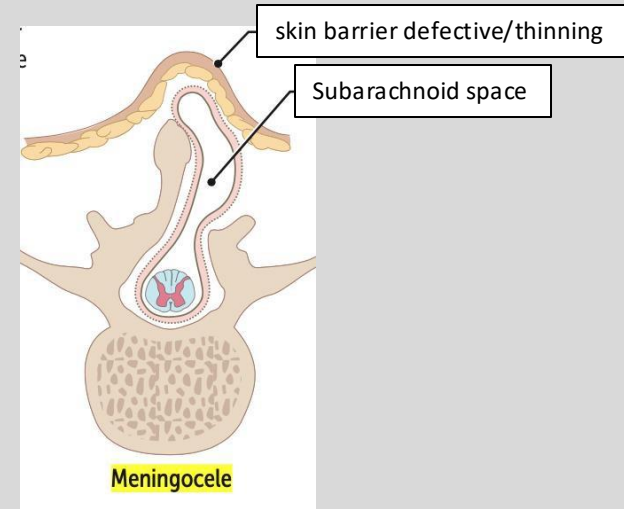
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<https://clinicalpub.com/polyuria-and-urinary-incontinence/>

Meningocele

- **Open/closed** neural tube defect (**but neural tube typically closes**)
- When the **meninges herniate** through a defect in posterior **vertebral arches** (can be small or large herniations of meninges) often in the **lower back**
- Spinal cord is usually normal and in correct location in canal
- **Most** are **covered** with skin and pose **no threat**
 - **Leaking CSF** or thin/ translucent skin indicates urgent **surgery**
- Need to perform urologic evaluation for **neurogenic bladder**
 - ~90% children with spina bifida spectrum (like meningocele) **incontinent of urine**

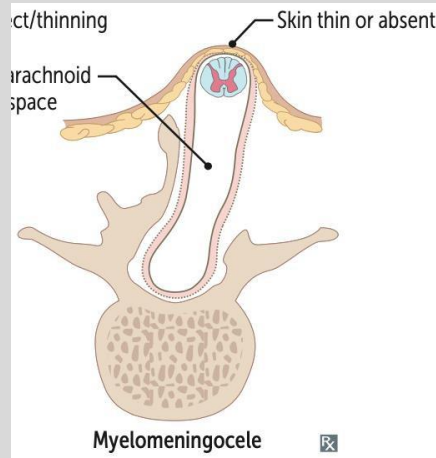


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Myelomeningocele & Myeloschisis

- **Open** neural tube defects (**failure of neural tube closure**)
- **Meninges & neural tissue herniate** through bony defect and are covered by thin skin an/or leptomeninges
- Almost uniformly associated with **Chiari II malformation**
- **Myeloschisis** (focused on spinal cord)/ **rachischisis** (spinal column focus): **Most severe form**, refers to **exposed & unfused neural tissue without skin/meningeal covering** – terms often used interchangeably



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- <https://clinicalgate.com/myelomeningocele-and-myelocystocele/>
- Myelomeningocele - central canal is seen going through, horizontally, the neural placode (failure of proper neural tube closure, leaves a flat plate of neural tissue called the neural placode)

Encephalocele and cranial-meningocele

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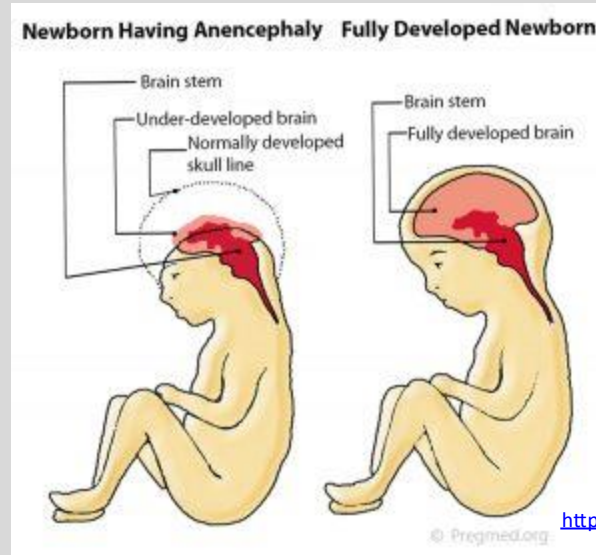
- Due to **protrusion** of tissue through **midline** cranial defect (cranium bifidum/ encephalocele = **skull doesn't close** completely during development)
- Cranial-meningocele = **CSF-filled meningeal sac only**
- Encephalocele = contains the **sac plus cerebral** cortex, cerebellum, or portions of the brainstem
- Associated with aqueduct stenosis, Chiari malformation, or Dandy-Walker, ↑ risk for developing hydrocephalus



<https://en.wikipedia.org/wiki/Encephalocele>

Anencephaly

- **Open** neural tube defect due to failure of **rostral neuropore** to close by wk. 4 (complete **failure of cephalad neural tube closure**) → persistent connection between amnion and spinal canal
- Results in **no** forebrain open calvarium, **cranial vault** is absent, skin is continuous with nervous tissue
- **Cerebral hemispheres & cerebellum** usually absent
- Only a **residue of brainstem** identified
- Will present with **↑ alpha-fetoprotein** & **polyhydramnios** (excessive amniotic fluid) because of fetuses non-intact swallowing centers
- **Incompatible with life**



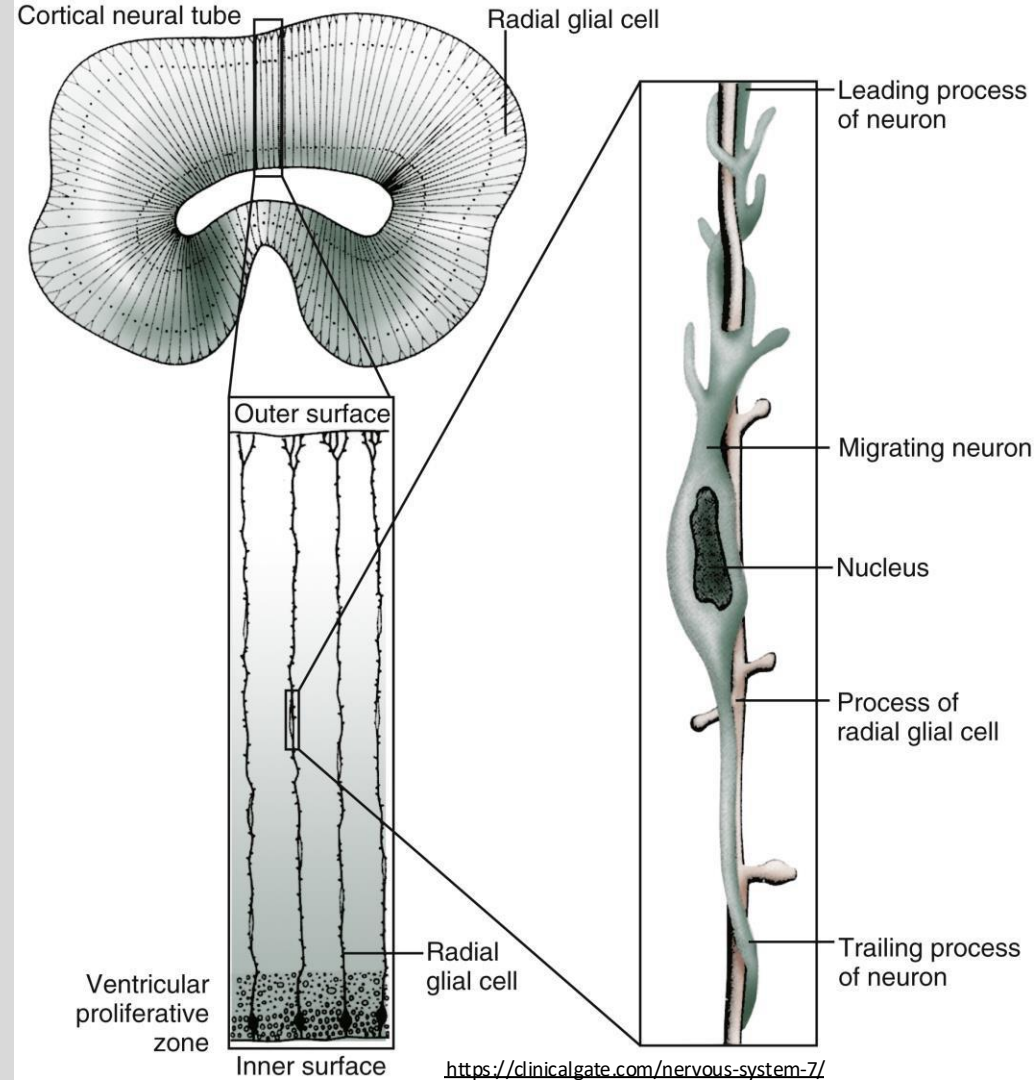
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Disorders of neuron migration

- Range from clinically silent minor abnormalities to devastating malformations
- Mechanisms: Poor control of **neuron** migration **via radial glial fiber system** that guides neurons to their proper site
- Normally migrating neurons attach to radial glial fiber, then **go to predetermined sites** to form 6- layered cerebral cortex



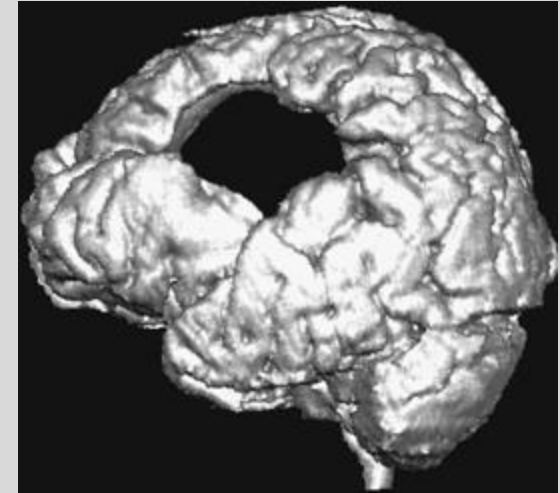
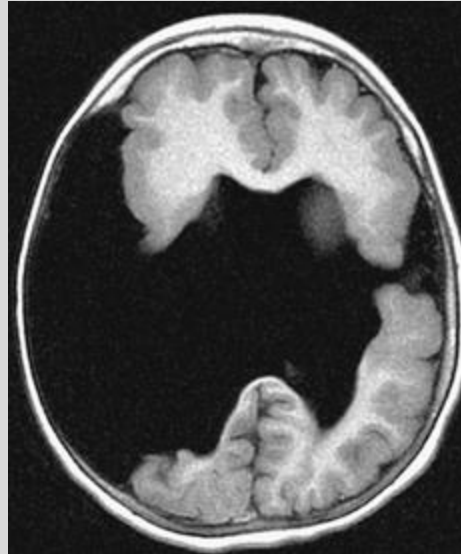
Lissencephaly

- AKA **agyria**, or “**smooth brain syndrome**”
- Absence of cerebral convolutions and a poorly formed sylvian fissure, giving **appearance of a 3- 4 month fetal brain at the time of birth**
- Associated with **failure to thrive**, **microcephaly**, CSF-related disorders (**hydrocephalus**), marked developmental **delay**, intellectual **disability**, and often a severe **seizure** disorder
- Ocular abnormalities are common, including hypoplasia of optic nerve and microphthalmia



Schizencephaly

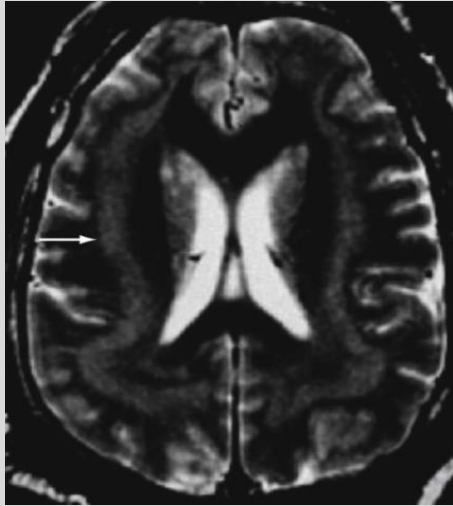
- “schiz-” = split
 - (ex. Schizophrenia = *splitting of the mind*)
- Presence of **clefts within hemispheres** due to abnormality of morphogenesis
- Frequently borders are surrounded by abnormal brain, particularly **microgyria**
- When bilateral, frequent severe **intellectual disability**, **intractable seizures**, **microcephaly** with **spastic quadriparesis**, and **hydrocephalus**
- If **anti-seizure drugs** fail, doctors might perform **surgery** to remove the part of the brain causing seizures (**epileptogenic tissue**)
- **Hydrocephalus** may require **surgery** for shunt placement



- <https://radiologykey.com/9-schizencephaly/>
- Axial T1 MRI left, surface rendering of left hemisphere

Neuronal heterotopias

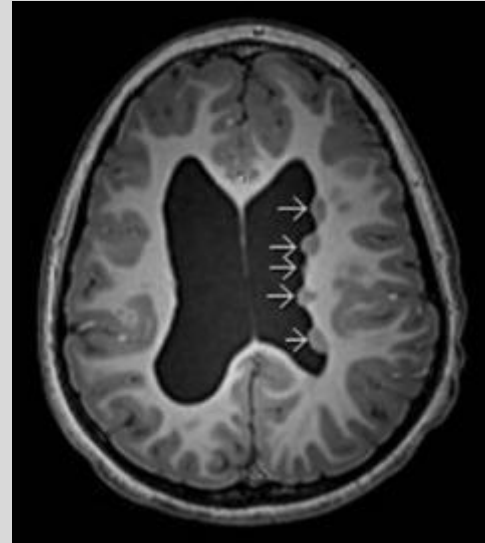
- Characterized by **altered gyral patterning/organization** or presence of **ectopic neurons in subcortical white matter or periventricular region**
- Several subtypes, main ones shown
- **Intractable seizures** are a common feature
- Several genes have been identified that are a cause of these conditions



Subcortical band heterotopia: Thin layer of gray-matter inside the white matter - **neurons did not migrate to surface**

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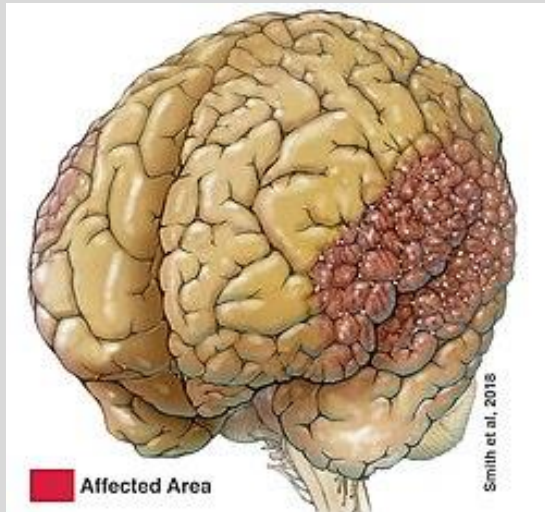
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- Periventricular nodular heterotopia: **Neurons did not migrate superficially from the edges of the neural tube**

Polymicrogyria

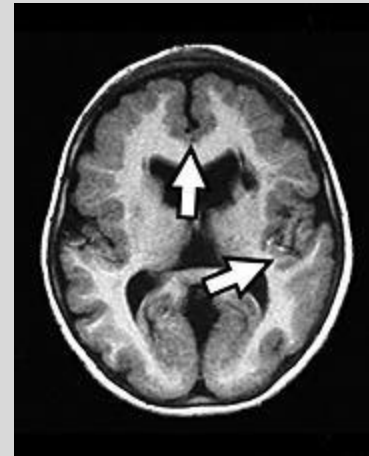
- Excessive number of **abnormally small gyri**
- Can be from **genetic mutations**, but also **prenatal infections** or **abnormal twinning** (ex. conjoined twins)
- Most frequent site is **perisylvian** (fissure)
- Clinically associated with drug-resistant **epilepsy**, upper motor neuron **paresis**, **cognitive impairment**



<https://en.wikipedia.org/wiki/Polymicrogyria>

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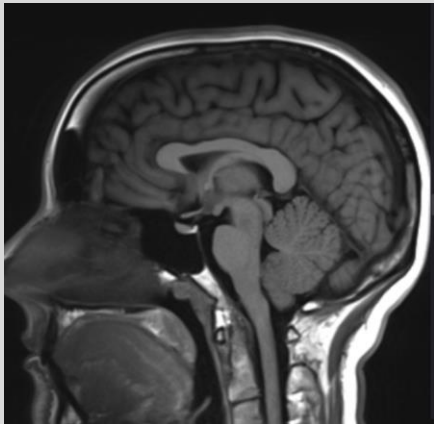
<https://candicelange.com/wp-content/uploads/2017/11/Figure-7-Human-symmetric-polymicrogyriaAxial-transverse-MRI-scans-of-patients-with.png.jpeg>

Agensis/dysgenesis disorders

- Brain developmental disorders due to **defective formation of the prosencephalon** and inadequate induction of forebrain structures
 - (Prosencephalon is the embryologic structure that develops into the diencephalon [**neocortex**] and telencephalon [**thalamus/hypothalamus**])

Corpus callosum agenesis

- Heterogeneous group of disorders that vary in expression
 - When an **isolated phenomenon**, the patient **may still be normal**
 - **When accompanied by brain anomalies** (heterotopias, polymicrogyria, etc.), patients often have significant **neurologic abnormalities**: Developmental delays (motor/speech), seizures, feeding issues, vision/hearing problems, poor coordination, and cognitive difficulties, with subtle social/abstract thinking issues later
- Multifactorial causes, from direct insult to developing structures or disruption of genetic signaling
- Can be **partial or complete** agenesis



MRI shows a **normal** sagittal head MRI

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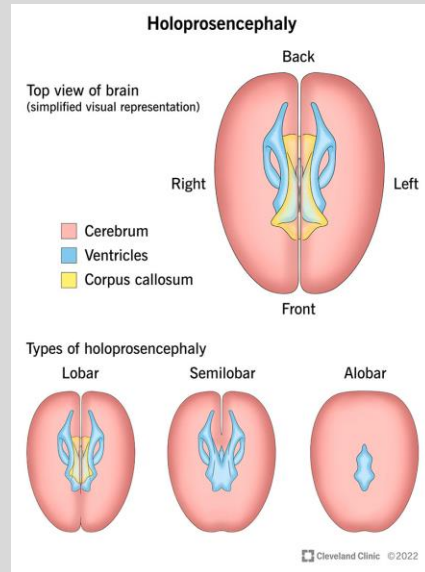
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- https://en.wikipedia.org/wiki/Agenesis_of_the_corpus_callosum
- MRI shows **complete agenesis** of the corpus callosum

Holoprosencephaly

- Failure in separation of **midline forebrain** at approximately 33 days' gestation
- Several subtypes: **alobar** (worst), **semilobar**, **lobar**
- Implicated with **Sonic Hedgehog (SHH)** pathway **mutations**
- ↑ risk for pituitary dysfunction (ex. **diabetes insipidus**), seen in **Patau syndrome** (trisomy 13)
- Accompanied spectrum of **craniofacial malformations**
 - (ex. **cyclopia**, iris colobomas, **cleft palate**, single central incisor tooth, single nostril)



- <https://www.semanticscholar.org/paper/Holoprosencephaly-with-cyclopia%3A-a-rare-case-report-Rathod-Samal/779a83250dce50b78bea72af6681d2f390537d15>
- Holoprosencephaly with cyclopia

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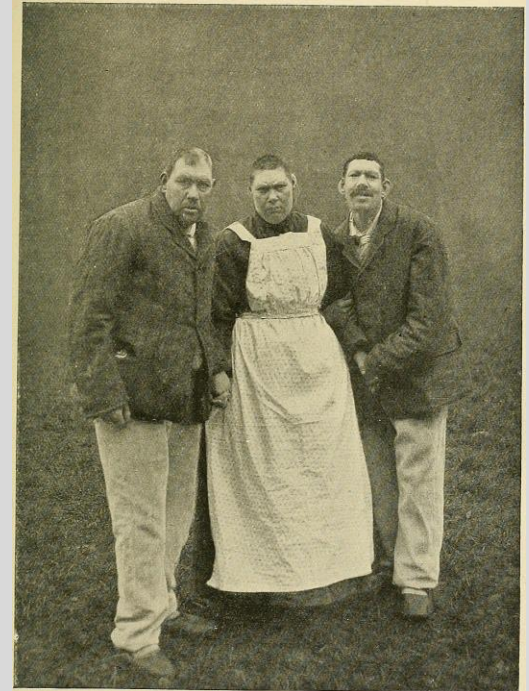
- <https://radiopaedia.org/articles/holoprosencephaly>
- Semilobar
- MRI reveals monoventricle

Microcephaly

- Abnormally **small head & brain** - head circumference that measures **>3 SD below the mean** for age and sex
- Many causes
 - Genetic:** Trisomy 21, Trisomy 18, Cri-du-chat
 - Infectious:** Zika virus, Congenital cytomegalovirus, rubella, toxoplasmosis, meningitis
 - Teratogenic:** Fetal alcohol syndrome, Fetal hydantoin (phenytoin-induced)
 - Other:** Radiation, malnutrition, hyperthermia
- In the United States, occurs in 1 in 800-5,000 births
- Commonly have **developmentally delayed** children, and as they grow often have an intellectual disability, poor motor function, poor speech, abnormal facial features, seizures and dwarfism
- In general, **life expectancy** for individuals with microcephaly is **reduced**, and **prognosis for normal brain function is poor**
- Occasional cases develop normal intelligence and grow normally (apart from persistently small head circumference)
- Often seen with other conditions** of cortical development

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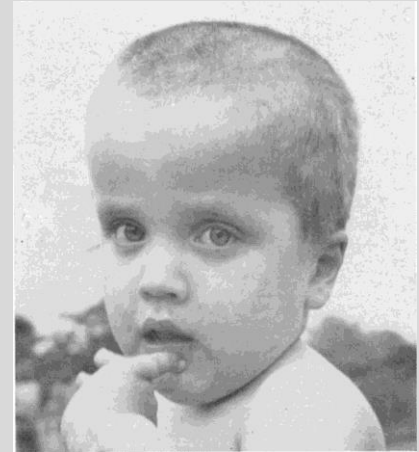
- <https://en.wikipedia.org/wiki/Microcephaly>
- 3 relatives, 1913

Megalencephaly

- Brain weight: Brain with an average weight that is 2.5 standard deviations above mean of the general population - often due to an increased number or size of brain cells
- Strong **link** between **genetic pathways** that cause **brain develop** and mutations in that pathway that result in **brain overgrowth**
- A mutation in PI3K-AKT pathway believed to be primary cause of brain proliferation
- The **most common cause is benign familial megalencephaly**: In some, brain is just large as a normal family trait (often inherited autosomal dominant) and **no other structural abnormalities or neurological issues**
- Affects 3–4 times more males than females
- Several neuropsychiatric disorders linked with megalencephaly, but studies have shown **autism** is most prevalent association
- Because the majority of megalencephaly cases are linked with **autism**, the **prognosis is equivalent** to the corresponding condition
- However, **macrocephaly** (megalencephaly is a type) is a known feature of more than 100 syndromes (ex. hydrocephalus)

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<https://thejns.org/fileasset/JNS300px.png>

Posterior fossa anomalies

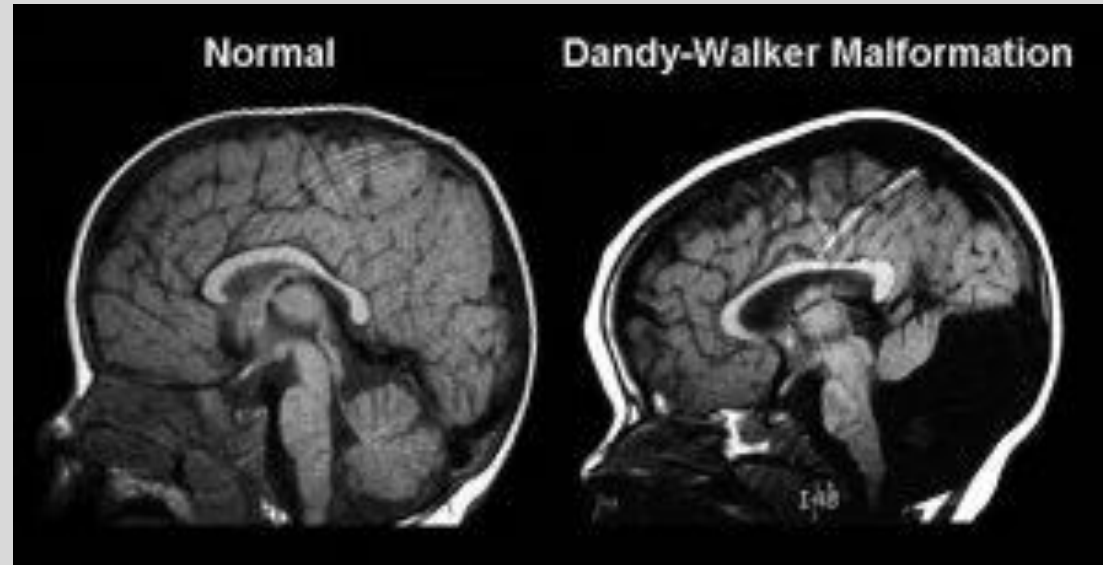
- Dandy-Walker
- Chiari Malformations

Dandy-Walker malformation

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- Many definitions with some disagreement, but generally:
- Presence of a **large posterior fossa cyst** communicating with **fourth ventricle**, complete or partial absence of cerebellar **vermis**, defect/displacement of the **cerebellar hemispheres**
- 4th ventricle (& other ventricles may be enlarged/ dilated with fluid - **hydrocephalus**) will be enlarged causing posterior fossa to enlarge too
- 1 out of every 5000 live born infants
- Mortality rate of 12.5% was observed. - most common cause of death is **complications from hydrocephalus** or its treatment



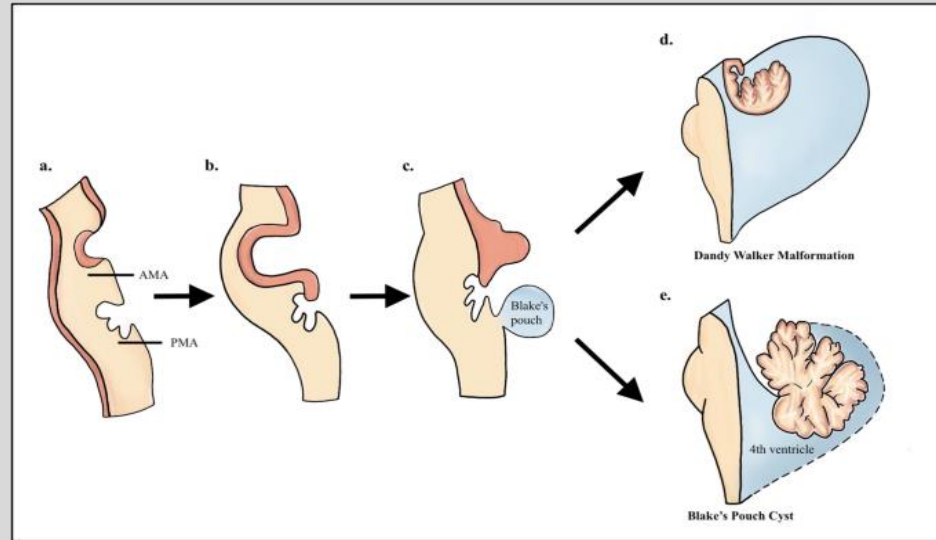
- <https://www.fetalhealthfoundation.org/fetal-syndromes/dandy-walker-malformation/>
- T1 MRIs

Clinical presentation of Dandy-Walker

- Usually presents in **1st year** with **hydrocephalus** & less severe cases are **often undiagnosed** until adolescence/adulthood
- Most common presentation is **macrocephaly**
- Other presentations: **seizures**, spasticity, lethargy, delayed milestones, **respiratory failure**, apnea, deafness, visual problems, increased intracranial pressure, **spina bifida** and **noncommunicating hydrocephalus**

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- <https://link.springer.com/article/10.1007/s00381-025-06842-0>
- Indentation is known as the plica choroidalis and divides roof plate into an upper portion called anterior membranous area (AMA) and a lower portion known as posterior membranous area (PMA)
- PMA fills with CSF, 4th ventricle balloons out forming the *Blake's pouch*, which later *perforates* and gives rise to median foramen of *Magendie*
- Blake's pouch cyst occurs when a normal embryonic *outpouching* of 4th ventricle *fails to regress* it persists as a 4th ventricle obstructing cyst causing cerebrospinal fluid (CSF) buildup and potential *hydrocephalus*

Chiari malformations

- A **collection** of hindbrain **abnormalities** ranging from simple **herniation** of cerebellar tonsils through foramen magnum to complete **agenesis** of cerebellum
- Likely multifactorial
- **Most symptomology** comes from **abnormal CSF dynamics** across cranio-cervical junction - guides treatment

Several types of Chiari malformations

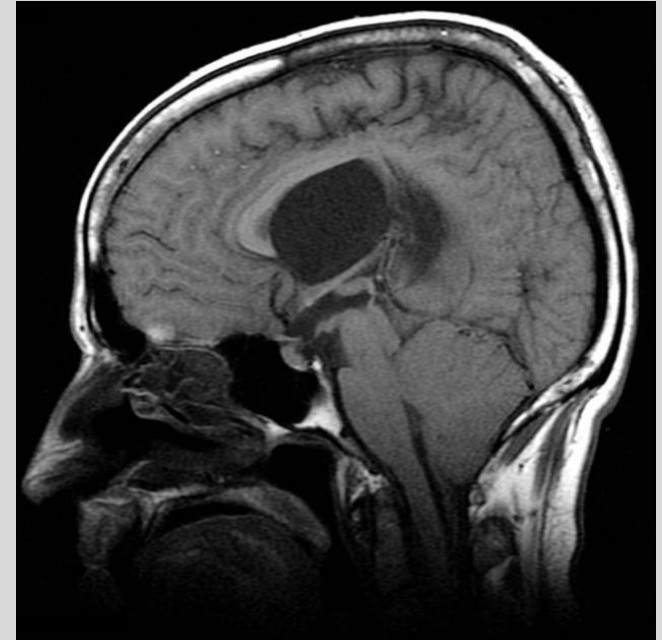
- The ones you need to know are Type I & Type II - Type II is 'Arnold-Chiari' malformation)
- The **Chiari I malformation** consists of ≥ 5 mm caudal displacement of the cerebellar **tonsils** inferior to the foramen magnum (top image, head T1 MRI)
- In **Chiari II** malformations, structures herniating through foramen magnum include cerebellar **vermis**, **brainstem**, **4th ventricle** (bottom image, head T1 MRI)

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- <https://radiopaedia.org/articles/chiari-1-malformation-1?lang=us>
- Chiari I malformation left
- Chiari II malformation right



Presentation of Chiari malformations

- Chiari I
 - Congenital, usually asymptomatic in **children** & manifests in **adults**
 - **Varies** from subclinical to **headaches** to severe **myelopathy & brainstem dysfunction**
 - **Most common** symptom is **occipital headaches**, which get **worse with Valsalva** maneuvers (laughing, coughing, sneezing) - these actions increase ICP because increased intrathoracic pressure decreases blood return from head (internal jugular vein) & transient venous brain congestion causes symptoms
 - Up to 70-80% have some degree of **ophthalmologic, otologic, or cerebellar dysfunction**
 - Associated with spinal cavitations (ex. **Syringomyelia** in a fluid filled cyst called a **syrinx**)
 - Lower **cranial nerve dysfunctions** are common (7-12)
 - Usually managed **conservatively** with **follow-up and imaging**
- Chiari II
 - Herniation through foramen magnum causes noncommunicating hydrocephalus due to aqueduct stenosis & they develop lumbosacral myelomeningocele
 - >95% of patients with **myelomeningocele** have CMII (**leading cause of death due to hydrocephalus**)
 - ~ 1/3rd develop **brainstem symptoms by age 5**, and > 1/3rd of those **die, usually of respiratory failure**
 - Usually presents as a neurosurgical **emergency** – goal = **relieve pressure from brainstem** and allow **CSF flow** to spinal canal

Other Chiari malformations

- **Chiari III** is **rare** and **most severe**, profound herniation and brainstem dysfunction, brain tissue protrudes through skull, **seldom compatible with life**
- **Chiari IV** includes **cerebellar hypoplasia + herniation**
- (There is also Chiari 0, syringomyelia with no tonsil herniation; and Chiari 1.5 is in between severity of type I and II classifications)

Conditions involving CSF

- Syrxinx/syringomyelia
- Hydrocephalus

Syrinx/syringomyelia

- CSF-filled **cavity** in spinal cord **central canal**
- Appears in **30-70%** of cases with **congenital Chiari I**
- Theories:
 - **4th ventricular outflow/ cervicomedullary junction obstruction**, forcing CSF down the spinal/ central/ **ependymal canal** creates a syrinx
- Also seen with **posterior fossa tumors** that obstruct CSF flow around foramen magnum
- Also seen with **trauma, infectious** processes, aneurysmal **subarachnoid** hemorrhages which can cause inflammation, injury and scarring that **can block CSF flow**
- Prolonged force of CSF overload in the canal causes expansion and syrinx formation
- **Syrinx** = tube, like a **syringe**

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- <https://radiopaedia.org/articles/syrinx-1?lang=us>
- Sagittal neck T1 MRI

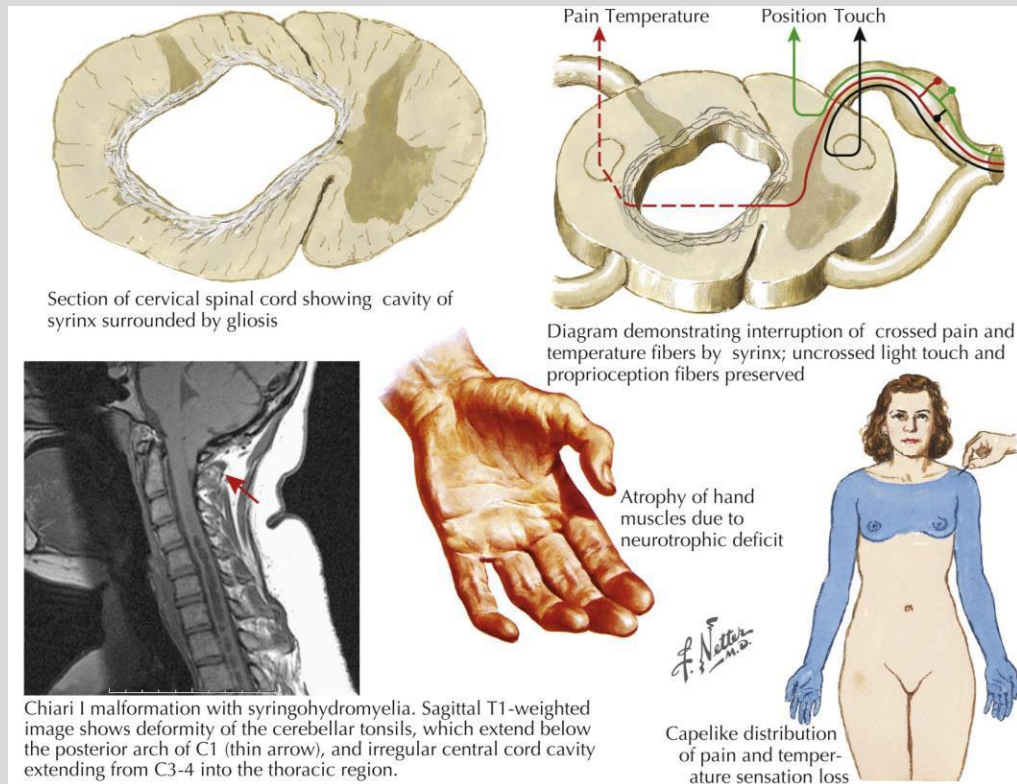
Clinical presentation of syrinx formation

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- Syrinx formation interrupts decussating (crossing midline) fibers of **spinothalamic tract**
- Disrupts **pain & temperature** sensation
- **Common** location of syrinx (**cervicothoracic**) usually yields **cape-like** distribution of **pain/temp disruption** in the **upper extremities**

• <https://www.netterimages.com/syringomyelia-unlabeled-neurology-neurosciences-frank-h-netter-5734.html>



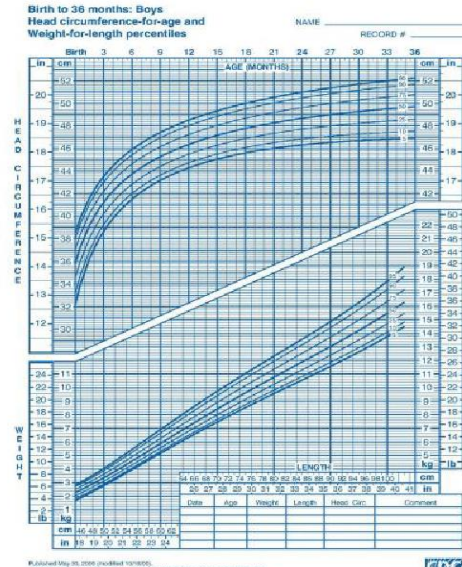
Hydrocephalus

- **Most common** condition treated by **pediatric neurosurgeons**
- Hydrocephalus is not a specific disease - a **diverse group** of conditions that result from **impaired circulation** and/or absorption of **CSF**
- **Congenital** causes include:
 - Aqueduct stenosis
 - Dandy-Walker
 - Holoprosencephaly
 - Myelomeningocele
 - Etc.
- **Acquired** Causes include:
 - Intraventricular hemorrhage
 - Brain tumors
 - Post-surgical (scarring/adhesions)
 - Post-traumatic

Clinical presentation of pediatric hydrocephalus

- Babies and young children have open skull sutures (soft spots or **fontanelles**) which are crucial for **brain growth** and increased head circumference - most closing gradually over the first couple of years, though some remain open until late childhood or adolescence
- Most important early finding in **head circumference** is **changing percentiles** over time (i.e. 50th -> 90th)
- Typical **Sx/Sx** are **↑ICP**, vomiting, extraocular movement lesions, dementia, ataxia, incontinence, papilledema
- Severe cases, profound **macrocephaly**

- https://www.researchgate.net/figure/Hydrocephalus-with-increased-head-circumference-in-a-3-month-old-child_fig4_234071019
- Increased head circumference in a 3-month-old child
- Pediatric head circumference chart from the CDC on the left



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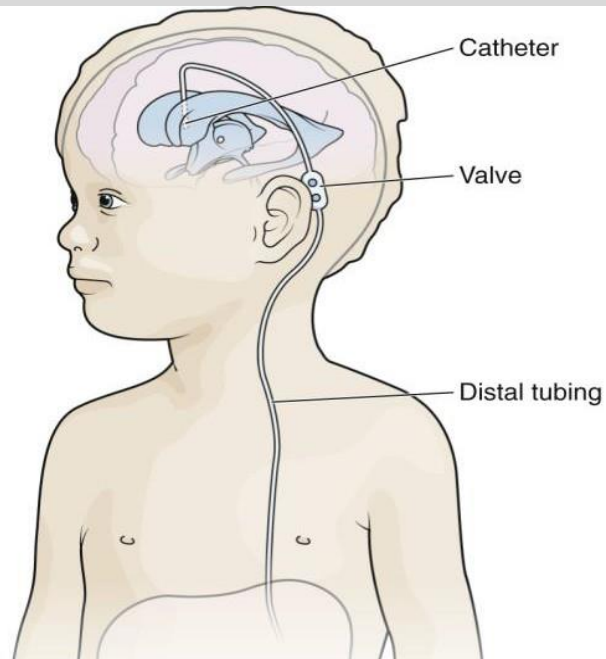


Management of hydrocephalus in pediatrics

- Usually **surgical**
- Most patients get a ventriculoperitoneal shunt
- A **VP shunt catheter** (ventriculoperitoneal) is usually placed in lateral **ventricle** via the scalp and in the **peritoneal cavity**
- Subcutaneous tube drains CSF from the ventricles to peritoneal cavity

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<https://clinicalgate.com/management-of-increased-intracranial-pressure-and-intracranial-shunts/>

- ventricular shunt system is composed of a ventricular catheter, valve mechanism, and distal tubing

Cerebral palsy

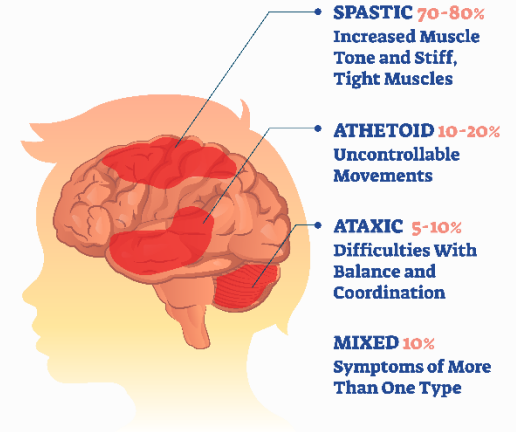
- A group of **disorders of movement development and posture** attributed to **non-progressive disturbances** that occurred in the **developing fetal or infant brain** (“acquired” happens in first month postnatally)
- Insult to CNS occurring (1) prenatally, (2) during birth, or (3) in the first 2 years of life
- 85-90% due to cerebral injury before or during birth
- Most common cause is perinatal **stroke**, can be **hemorrhagic, ischemic, or thromboembolic**
- 2nd cause is infantile **meningitis** and/or **encephalitis**
- Also happens due to trauma (**abuse, MVAs**)
- **Premature** birth is a **leading risk factor** for cerebral palsy (CP)



Diagnosis and presentation of CP

- In infants, **primitive reflexes** that have not yet integrated by usual period may be an early diagnostic clue
- **Intellect & cognition** are **intact**, as this is a condition of the **motor system**
- May adopt certain **postures**
 - **Arms** (e.g., flexion of elbows, wrists, and fingers, with internal rotation of the shoulders and pronation of the forearms)
 - **Legs** (e.g., flexion of the hips, knees, and toes, with adduction of the hips).
- **Spasticity**
- Associated with various movement disorders (**dystonia, ataxia, choreoathetosis**)
- **Imaging** is usually **normal**, in some cases there is **subtle** loss of volume or MRI hyperintensity in certain regions (i.e. motor cortex)

Types of Cerebral Palsy



Thank you!

Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>